Field	Entry
Source ID	WBC-0010
Title	Microstructure and Mechanism of Grain Raising in Wood
Source type	Paper (open access)
Link / File	DOI: 10.3390/coatings7090135
Craft applicability	Wood
Primary Why-theme	11. Product Function, Use and Performance (with secondary link to 10. Failure, Repair and Quality)
Process stage	Finishing (sanding) / product use (wetting)
Questions answered / generated	Why does a sanded wood surface become rough again after wetting? Why are some wood species more prone to grain raising than others?
Evidence level	E3 (direct scientific evidence, profilometry and SEM characterization)
Relevance note	Shows that sanding tears cell walls to create loosely bonded fibre slivers that project after wetting, and that grain raising is inversely related to wood density — directly explaining why finished Udayagiri surfaces (spoons, bowls, handles) may roughen again after washing, and suggesting species density as a relevant factor for food-contact product performance.